



COMPREHENSIVE USER GUIDE

CorpusMind

Local-first, AI-native research environment
for corpus linguistics and multimodal discourse analysis

ENGINE

Python + FastAPI + spaCy

PLATFORMS

Win / Mac / Linux / PWA

LICENSE

AGPL-3.0

v1.2.6

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PART 1

Citation

If you use CorpusMind in your research, please cite it as:

Mandour, W., & Ibrahim, W. (2026). **CorpusMind: A local-first, AI-native research environment for corpus linguistics and multimodal discourse analysis** (Version 1.2.6) [Computer software]. Zenodo. <https://doi.org/10.5281/zenodo.21226650>

PART 2

1. What is CorpusMind?

CorpusMind is a research tool for corpus linguists and discourse analysts. It lets you upload texts, run concordance searches, compute collocations, keyness, dispersion, and vocabulary profiles, analyse grammar and discourse features under named academic taxonomies, research learner language, and work with Arabic corpora using CAMEL Tools. A Vision Suite (and the companion **CorpusMind Lens** app) extends the same workflow to images using Kress and van Leeuwen's Visual Grammar and eight discourse lenses.

The AI Assistant answers questions about your corpus by calling the analysis tools and citing specific evidence (concordance line IDs, computed statistics). Every answer is either **grounded** (backed by a tool call) or clearly flagged as **ungrounded** — never silently presented as fact. A floating version of the same assistant is available from every analysis screen.

Everything runs on your own machine. No data leaves your computer unless you explicitly choose to use a cloud AI provider or the optional Gemini error-interpretation feature.

PART 3

2. Installation

Option A: Desktop App (recommended)

Download the installer for your platform from the [releases page](#): Windows (EXE/MSI), macOS Apple Silicon and Intel (DMG), Linux (AppImage/deb). Install [Ollama](#) as well — the app starts it automatically and downloads recommended models with one click from **Settings** → **Model Providers**.

The desktop app automatically:

- bullet Starts the Python engine in the background (and waits until it is healthy)
- bullet Starts Ollama in the background (if installed)
- bullet Restarts either backend if it goes down while the app is running (v1.2.6 self-heal)

bullet On macOS, closing the window no longer stops the backends (v1.2.5)

Option B: Development Mode

You need **Python 3.12**, **Node.js 20**, and **Ollama**.

```
ollama pull llama3.2:3b
```

Set up the engine:

```
cd CorpusMind/engine python3 -m venv .venv source .venv/bin/activate pip
install -e ".[dev]" python -m spacy download en_core_web_sm corpusmind-engine
```

Set up the web app (new terminal):

```
cd CorpusMind/web npm install npm run dev
```

Open `http://localhost:5173`. The engine listens on `127.0.0.1:8765` (override with `CORPUSMIND_PORT` / `CORPUSMIND_HOST`).

Optional: Arabic Support

```
pip install camel-tools pyrsistent muddler cachetools emoji future regex
camel_data -i morphology-db-msa-r13 camel_data -i dialectid-model6
```

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3. Creating a Project and Uploading Texts

1. Click **Corpora Selection** → **Your Corpus** in the sidebar.
2. Select an existing project from the dropdown, or click + **New Project**.
3. Click + **New** under Corpora to create a corpus (name, language, genre).
4. Click your corpus to select it.
5. Click the upload area (or drag and drop) to add text files.

Supported formats: TXT, DOCX, PDF, HTML, XML, CSV, Markdown. The engine detects encoding, cleans the text, tokenizes, tags POS, lemmatizes, and parses dependencies. For Arabic corpora it uses CAMEL Tools (if installed) for morphology, root extraction, and dialect identification. The pipeline recipe (model names and versions) is visible and exportable.

Corpus Cleaning

Each corpus has a **Clean** button with 16 options: collapse whitespace, remove URLs/emails, lowercase, remove punctuation/numbers/emoji, remove stopwords, minimum token length, and Arabic-specific normalization (alef variants, diacritics, tatweel). Cleaning is destructive — it re-processes every document.

Reference Corpus

Corpora Selection → **Reference Corpus** offers three options: **Download** (HuggingFace + Wikipedia + OPUS), **Upload** your own, or the **bundled** BE06 frequency list.

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4. Concordance Search (KWIC)

1. Select a corpus and open **Concordance**.
2. Type a query (* works as a wildcard); choose word, lemma, or POS level.
3. Set the context window and click **Search**.

Each line has a stable ID (`doc:0:3`) that the AI Assistant can cite as evidence. Results export to five formats.

Vector KWIC (Semantic Search)

Vector KWIC finds lines by *meaning*, not just form: pick an embedding model (`bge-m3`, multilingual, or `nomic-embed-text`, English), download it with one click, and search with natural-language queries (e.g. *"sentences about economic hardship"*). Ranking is raw cosine similarity over a local SQLite vector cache — no external API. On CPU-only machines the first search embeds the corpus in small batches and can take a while; see Troubleshooting §13.4.

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5. Frequency, Collocation, Keyness, and Dispersion

Frequency: word/lemma/POS lists with per-million and percent; STTR at the top.

Collocation: node word + window size; all 7 measures (MI, T-score, log-likelihood, Dice, LogDice, chi-square, Delta P) plus the interactive collocation network graph (exportable as SVG/PNG/JSON).

Keyness: target vs reference corpus; significance (log-likelihood, chi-square) beside effect size (Log Ratio, %DIFF, Simple Maths, Odds Ratio). **Methods PDF** auto-drafts a methodology paragraph for your manuscript.

Dispersion: Juilland's D and Gries' DP across documents.

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6. Advanced Analysis Tools

N-grams: n = 2–10 with Biber et al.-style frequency-and-range criteria for lexical bundles.

POS Analysis: distributions and POS n-grams; switch tagsets (UPOS, Penn Treebank, CLAWS7, CAMEL for Arabic) from Settings.

Grammar: passive voice, modals, negation, relative clauses, complex NPs, and tense from the dependency parse.

Dependency: query any UD relation (nsubj, obj, obl, ...) for common governor–dependent pairs.

Discourse: a user-selectable, citable taxonomy of discursive features:

Hyland (2005) metadiscourse — interactive (transitions, frame markers, endophorics, evidentials, code glosses) and interactional (hedges, boosters, attitude markers, self-mentions, engagement) categories.

Halliday & Hasan (1976) cohesion — reference pronouns, the four conjunction classes, and lexical repetition chains across adjacent sentences (substitution/ellipsis are not covered).

Martin & White (2005) Appraisal — engagement (entertain, attribute, deny, counter, proclaim), graduation-force intensifiers, and an inscribed affect starter set (invoked attitude is not covered).

CLAWS/USAS semantic tagset (top-level) — the bundled USAS lexicon maps every word onto one of 24 semantic categories (communication, cognition, emotion, politics...), each annotated with its discourse-functional group. Lexicon-based lookup — where the lexicon has no entry, tokens are honestly reported as *unmatched*; this is not the licensed CLAWS/USAS tagger.

Every result names and cites its taxonomy, so findings are reportable and comparable across studies.

Vocabulary: frequency bands (K1, K2–K9, AWL, Off-list), rare and academic words.

Sentiment: per-sentence scores (–1 to +1) with a timeline.

Metaphor: verb-based metaphor *candidates*; LLM triage and human verification required before counting any as confirmed.

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7. Learner Research Suite

Purpose-built for learner-corpus studies (Granger 1998; Housen & Kuiken 2009):

CAF Report: complexity, accuracy, and fluency indicators for the active corpus, including HD-D (heterogeneous hapax-based diversity).

CIA Compare: compare two corpora (e.g. learner vs expert writing)

across the CAF battery with effect sizes.

bullet **Error Patterns:** candidate learner-error patterns surfaced from POS

and dependency signals, ranked for manual inspection — candidates, not verdicts.

bullet **AI-vs-learner comparator:** inspect how AI-generated text differs

from learner text on the same indicators (research ethics: disclose AI use; see the Assistant's audit trail).

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8. Arabic Analysis

Arabic Tools in the sidebar:

bullet **Morphology:** root (al-jizr), pattern (al-wazn), lemma, POS, stem,

Buckwalter transliteration, number, gender, broken plurals (CAMEL calima-msa-r13).

bullet **Roots:** all words sharing a triliteral root (k.t.b → kitab, maktaba, katib, yaktub).

bullet **Dialect ID:** MSA, Egyptian, Gulf, or Levantine (CAMEL DIDModel6).

bullet **Buckwalter / Dediactize / Normalize:** script utilities.

bullet **Register:** Classical / MSA / Dialectal detection.

bullet **Translate:** Arabic–English lookup equivalents.

Arabic normalization is also available inside corpus cleaning, and Arabic is supported end-to-end in Vector KWIC via the multilingual bge-m3 model.

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9. Vision Suite and CorpusMind Lens (Multimodal Analysis)

Images are treated as corpus documents with metadata, provenance, and query tools, analysed against explicit frameworks. The same suite lives in the main app and in the dedicated **CorpusMind Lens** app.

9.1 Building an image corpus

Create a project and corpus, then in **Your Corpus** (Lens) or the **Vision Suite** (main app) create an **image set**, document provenance (source, period, selection criteria), and drag images in (PNG, JPEG, WebP, GIF, TIFF, BMP; ≤ 25 MB each, 50 per batch). Upload extracts OCR text, colour and composition features, and EXIF/XMP metadata — GPS coordinates are deliberately never extracted.

9.2 Metadata and per-image analysis

Each image carries IPTC-Core-aligned fields (source, date, licence, genre, language, notes); **Tag All Images** applies fields set-wide. Per image:

bullet **Analyse** — colour, composition (information value, salience, balance, vectors), OCR.

bullet **Visual Grammar** — Kress & van Leeuwen (2006) metafunctions, each claim a scored hypothesis.

bullet **Align** — image-region ↔ text-span alignment with confidence.

bullet **Discourse lenses** — 8 frameworks: Social Semiotic, CDA (Fairclough, van Dijk, Wodak, Machin & Mayr), Persuasion, Framing, Narrative, Visual Metaphor, Emotion, Cultural. Provenance badges show heuristic vs vision-LM mode.

bullet **Facial analysis** (opt-in, off by default) — descriptive cues only, never identity recognition.

9.3 Set-level tools

Vision-LM descriptions with a local vision model, recurring-theme aggregation, OCR corpus tools (KWIC over OCR text, frequency lists, set-vs- set keyness by log-likelihood), spreadsheet export with provenance, and **Export OCR corpus** — the set's text as a <doc>-marked corpus file for the main text tools.

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10. The AI Assistant

AI Assistant in the sidebar opens a tool-using agent (not a chatbot): it selects and calls analysis tools, then answers from their output. Tool use **grounded** (green badge); otherwise **ungrounded** (orange). A **floating assistant** button (bottom-right) offers the same grounded chat from every analysis screen, with context about the view you are on.

Confidence layer: the Assistant self-assesses confidence; below 70% you answer 2–3 verification MCQs before the answer is revealed.

Human verification: Accept / Reject / Edit every answer; the decision is recorded in the audit trail and the AI-usage disclosure for your Methods section.

Student mode (Settings → Research & Reproducibility): hides the AI answer until the student writes their own interpretation.

Example questions: *"What are the top 10 content words?"*; *"Find 'however' in context"*; *"What are the strongest collocates of 'patient'?"*; *"What hedges does this author use?"*; *"Compare this corpus against the reference."*

PART 12

11. Export, Collaboration, and Privacy

Export: every analysis exports as Excel (`.xlsx`), CSV, TSV, plain text, or JSON via the Export dropdown. The Collocation network exports as SVG/PNG/JSON. **Methods PDF** (Keyness view) drafts the methodology paragraph with exact tools, versions, and formulas.

Corpus Hub: the Reference Corpus view searches open-access corpora — HuggingFace Datasets, live Wikipedia (CC-BY-SA), and 1,200+ OPUS parallel corpora. Downloads land in your browser; searches are proxied through the engine, so your own data never leaves the machine.

Saved searches and bookmarks: save queries with parameters; bookmark lines/statistics with notes.

Project sharing: mark a project shared (public/private) with a share token; sync events are audited.

At-rest encryption: optional AES-256-GCM for image files via `CORPUSMIND_ENCRYPTION_KEY` (the key is never stored on disk).

Accessibility: WCAG 2.1 AA target — focus indicators, skip link, high contrast mode, reduced motion, 44px touch targets, full RTL mirroring.

Smart Troubleshooting: backend errors appear in the taskbar. With a Gemini API key configured (Settings → Smart Troubleshooting, off by default), errors are auto-interpreted in plain language with a likely cause and suggested fix; error context leaves the device only with your explicit acknowledgment. **Report to developer** opens a pre-filled email.

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12. PWA versus Desktop Application

Both forms share the same UI, project format, and engine. The **PWA** (hosted at `https://corpus-mind-web.vercel.app/`) runs in the browser — installable, offline-capable after first visit, with the engine started by you or on a lab server. The **desktop application** (Tauri 2, Windows/macOS/Linux) bundles UI + supervisor + engine sidecar: the supervisor spawns and health-waits the engine, auto-starts Ollama, self-heals backends, and writes `engine.stdout.log` /

`engine.stderr.log` to the OS log directory.

Capability	PWA	Desktop
Analytical tools & Vision Suite	Identical	Identical
Engine delivery	Manual start / lab server	Automatic (supervised, self-healing)
Ollama	Run locally yourself	Auto-spawned if on PATH
Offline	Shell cached; engine needed for analysis	Fully offline
File access	Browser dialog uploads	Native dialogs + data dir
Logs	Managed by you	Managed for you
Data residency	Local or lab server	Always local
Updates	Automatic on next visit	New installer
Best for	Trying it out, lab-server teams, locked-down machines	Sensitive corpora, classrooms, full offline

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13. Troubleshooting Common Issues

Most issues are environmental (a backend that is down, a missing model, or another application interfering with local connections) rather than bugs. Work through this list before reporting a problem; the Smart Troubleshooting panel (taskbar → error → Details) may already name the cause.

13.1 Security or grammar software intercepting Ollama (HTTP 500)

Symptom: "HTTP 500 Internal Server Error" when chatting or running AI features, often on a fresh installation, even though Ollama is installed and the model exists.

Known cause: desktop applications that inspect or proxy local HTTP traffic can intercept CorpusMind's requests to Ollama (`127.0.0.1:11434`). Confirmed real-world case: **Grammarly** running in the background caused exactly this; quitting it made the errors stop. Antivirus "web shields", VPNs, and corporate proxies can do the same.

Fix: quit or temporarily disable the interfering application (e.g. right-click the Grammarly tray icon → Quit), or add CorpusMind and Ollama to its exclusions/allow-list, then retry. If your organisation manages the machine, ask IT to exempt loopback (`127.0.0.1`) traffic from inspection.

13.2 "Ollama is not running" (503)

The engine cannot reach Ollama on `127.0.0.1:11434`. Start Ollama (the desktop app tries to start it automatically), verify with `curl http://127.0.0.1:11434/api/tags`, and make sure at least one chat model is pulled (Settings → Model Providers offers one-click downloads). In browser/PWA mode you must start both the engine (`corpusmind-engine`) and Ollama yourself.

13.3 "Model not found — run ollama pull" (409)

The selected model is not installed. Download it in **Settings** → **Model Providers** (one click) or run `ollama pull <model>`. If the model is installed, update to the latest release — v1.2.2 fixed a tag-matching bug (`bge-m3` vs `bge-m3:latest`) that caused a false 409 after a successful download; no re-download is needed.

13.4 Vector KWIC is slow or times out (CPU-only machines)

Embedding a large corpus on CPU takes minutes of compute. Since v1.2.5 requests are chunked into small batches (override with `CORPUSMIND_EMBED_BATCH`), and since v1.2.4 the timeout is configurable (`CORPUSMIND_EMBED_TIMEOUT_S`). Warm the model first (Warm-up button), expect the first search to be the slowest (the vector cache persists), and prefer the smaller `nomic-embed-text` model for English-only corpora. If the engine reports `502 embedding_unreachable`, the Ollama connection dropped mid-request — check §13.1/§13.2 and retry; the desktop app will also offer to restart the backend (v1.2.6).

13.5 The engine is not reachable (PWA / manual mode)

Start it with `corpusmind-engine` and check `http://127.0.0.1:8765/api/v1/health`. If the port is busy, set `CORPUSMIND_PORT`. If the firewall prompted on first run, allow the engine. Desktop builds health-wait for up to 120 s at boot — on very slow disks or with aggressive antivirus, first launch can take that long.

13.6 macOS: window closed and now nothing responds

Since v1.2.5 closing the window keeps the engine and Ollama alive; click the dock icon to reopen. If a backend actually died, the app probes and restarts only what is down when you reopen (v1.2.6). If you are on an older version, closing the window stopped the backends — quit and relaunch instead.

13.7 Arabic tools are missing or fail

CAMeL Tools is optional. Install it (see §2) and download the two data files. If only some tools fail, check the engine log for the specific model (e.g. the dialect-ID model) and run the matching `camel_data -i` command.

13.8 Where are the logs? How do I report a problem?

Desktop logs live in the OS log directory as `engine.stdout.log` and `engine.stderr.log` (macOS: `~/Library/Logs/CorpusMind`; Windows: `%APPDATA%\CorpusMind\logs`; Linux: `~/.local/share/CorpusMind/logs` — names per platform convention). The taskbar error panel and **Settings** → **Diagnostics** (Run Diagnostics) summarise backend health. Use **Report to developer** in the error panel to open a pre-filled email to w.abumandour@squ.edu.om, or file an issue at <https://github.com/waleedmandour/CorpusMind/issues> with the log excerpt and your OS, app version, and Ollama model list.

14. CorpusMind Compared with Other Corpus Tools

CorpusMind complements rather than replaces existing tools. Claims below are sourced to each tool's own documentation/publications (2025).

AntConc (Anthony, free, desktop) — excellent for teaching and quick KWIC on pre-processed corpora, but has no POS tagger, lemmatiser, or parser, one user-selected collocation statistic, and no Arabic morphology, AI assistant, or multimodal support. CorpusMind bundles the full annotation pipeline and the seven-measure collocation suite with provenance records.

Sketch Engine (Lexical Computing, subscription, cloud) — the reference implementation of Word Sketches with ready-made 100+ language corpora. Cloud processing makes it unsuitable for confidential corpora without a data-processing agreement. CorpusMind is local-first and free, but does not (yet) implement Word Sketches.

#LancsBox (Lancaster, free, desktop) — strong visualisation (Graph Coll) and BNC/BNC64 support; lighter on statistics and without Arabic morphology or AI assistance. CorpusMind implements the full seven-measure collocation and four-measure dispersion suites (Juilland's D, DP, ARF, AWT) with provenance.

Voyant Tools (Sinclair & Rockwell, free, web) — fast visual exploration for digital humanities; raw-token only, no standard collocation statistics or annotation. CorpusMind adds linguistic annotation, formal statistics, and reproducibility.

Feature	CorpusMind	AntConc	Sketch Engine	#LancsBox	Voyant
Licence	AGPL-3.0 (free)	Freeware	Subscription	Free (academic)	Open source
Deployment	Desktop + PWA (local-first)	Desktop	Cloud	Desktop	Web
Collocation measures	7	1	Multiple (Word Sketch)	Several	1 (raw)
Keyness (LL + effect sizes)	Yes	LL, chi-square	Yes	Yes	No
Dispersion	D, DP, ARF, AWT	No	Yes	Visual	No
POS / lemma / dependency	Yes (bundled)	No	Yes	Limited	No
Arabic morphology	Yes (CAMEL)	No	Yes	No	No
Multimodal (images)	Yes	No	No	No	No
Grounded AI assistant	Yes (local LLM)	No	No	No	No
Provenance per operation	Yes (YAML)	No	Limited	No	No
Data residency	Local by default	Local	Cloud	Local	Cloud

Rule of thumb: Voyant for a quick visual overview; AntConc/#LancsBox for teaching and quick searches; Sketch Engine for Word Sketches and ready-made corpora; CorpusMind for reproducible,

framework-grounded, local-first analysis of text and images.

PART 16

Getting Help

- bullet GitHub: <https://github.com/waleedmandour/CorpusMind/issues>
- bullet Live PWA: <https://corpus-mind-web.vercel.app/>
- bullet Build Guide: docs/BUILD_GUIDE.md
- bullet Methodology Reference: docs/METHODOLOGY.md
- bullet Zenodo DOI: <https://doi.org/10.5281/zenodo.21226650>